ECE 590 – Fundamentals of Cloud Computing, Spring 2026

Hadoop Homework

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Github link: https://github.com/Aadityaa2004/Financial-Investment-Risk-Analysis/tree/main

Project Introduction:

This project is a distributed, cloud-native Hadoop pipeline deployed on a three-node AWS EC2 cluster (one master, two workers). The goal is to analyze a large Lending Club dataset (~500 MB) to compute **risk insights** across key segments like:

1) loan grade
2) State
3) employment history
4) interest rate

The architecture uses **HDFS** for distributed storage and YARN for resource-managed execution via **MapReduce**. Go microservices are used for REST API endpoints.

Data Pipeline:

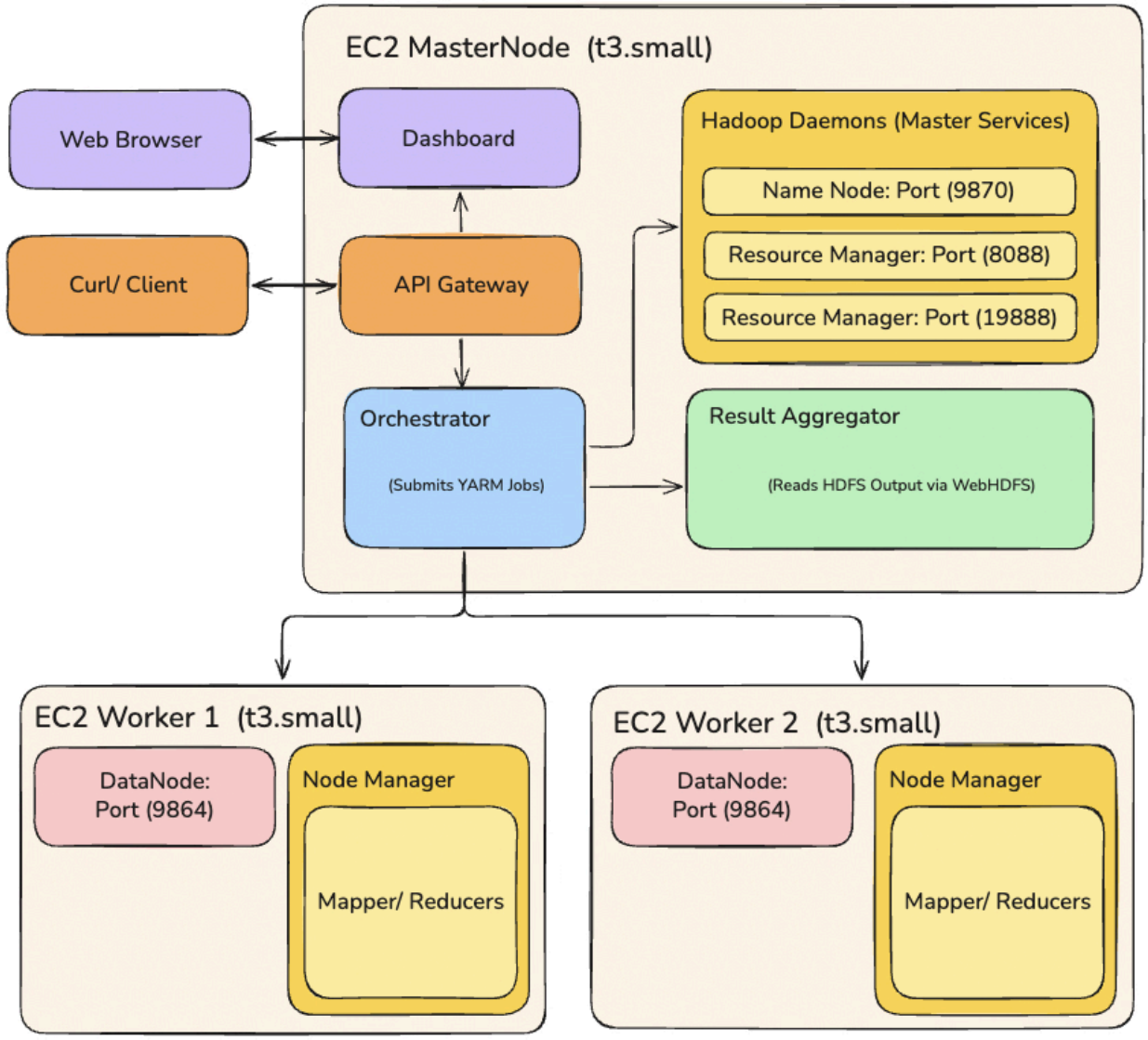
This project runs Four MapReduce jobs:

- **Job 1 (Grade Analysis):** Computes total loans, defaults, default rate, and risk level by loan grade (A–G).
- **Job 2 (State Analysis):** Calculates state-wise default behavior to identify high-risk geographies.
- **Job 3 (Employment Analysis):** Groups borrowers by employment length and evaluates default risk by bucket.
- **Job 4 (Interest Analysis):** Links interest rates with defaults and grade-level risk behavior.

Result Aggregation

- The reducer outputs from all four jobs are written back to **HDFS**.
- Aggregator logic combines the individual job outputs into a **Risk Summary**, which includes the overall default rate, identification of the highest/lowest risk groups, and final recommendations.

Architecture:



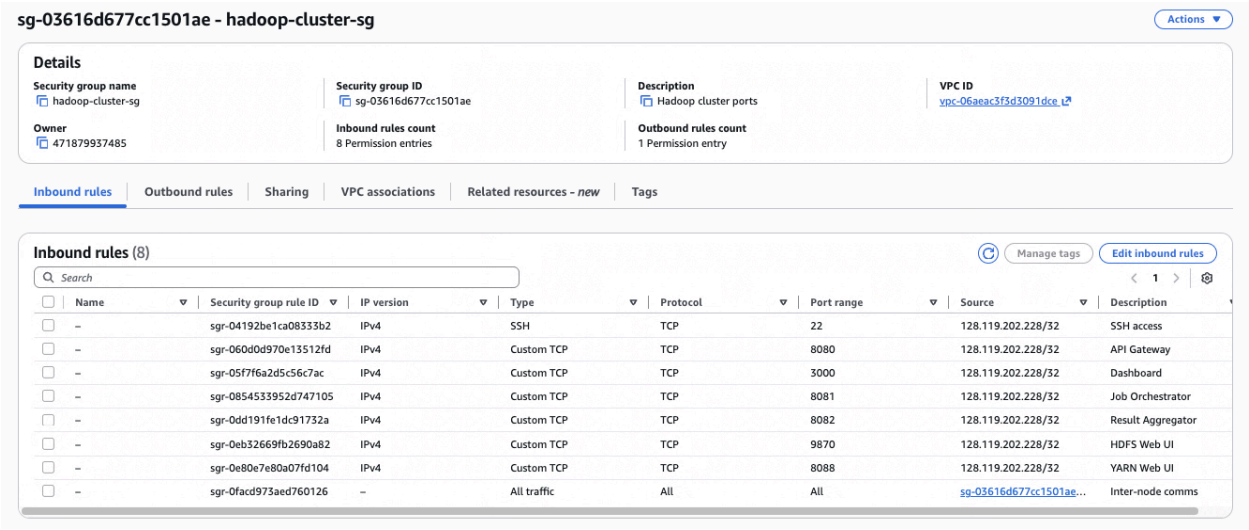
Setup Guide:

Sign into AWS, and generate a SSH key pair, and pick “us-east-1” region

STEP 1: Create a Security Group (Firewall Rules):

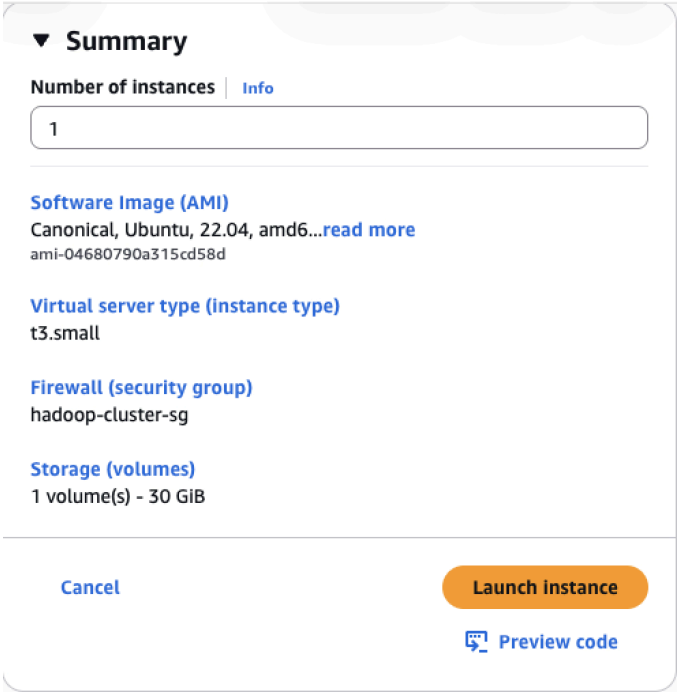
Here is the firewall rules table to insert above the screenshot:

Type	Protocol	Port Range	Source	Description
SSH	TCP	22	My IP	SSH access
Custom TCP	TCP	3000	My IP	Dashboard
Custom TCP	TCP	8080	My IP	API Gateway
Custom TCP	TCP	8081	My IP	Job Orchestrator
Custom TCP	TCP	8082	My IP	Result Aggregator
Custom TCP	TCP	8088	My IP	YARN Web UI
Custom TCP	TCP	9870	My IP	HDFS Web UI
All traffic	All	All	Custom →…	Inter-node comms

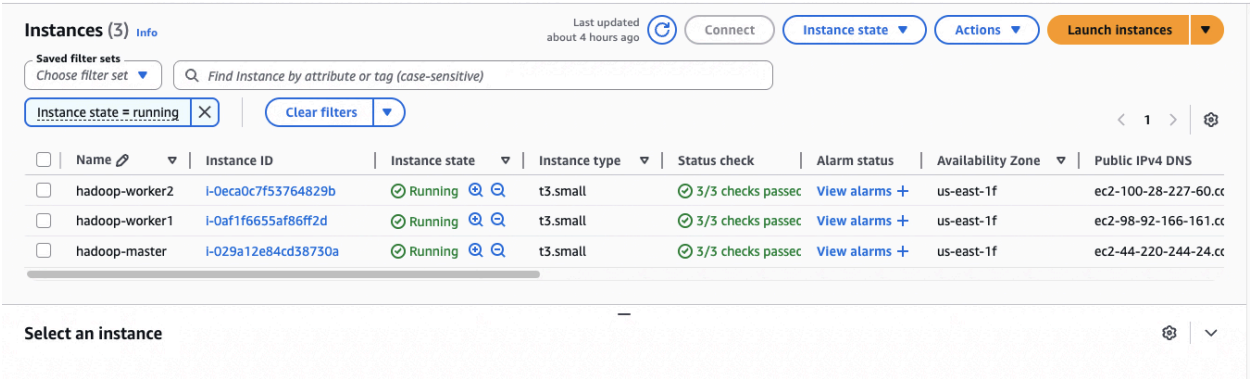


STEP 2: Launch 3 EC2 Instances:

Using the t3.small, create 3 Ubuntu 22.04 LTS servers, with storage volume 30GiB.



Once they have all been create, We can see the active EC2 instances in the dashboard:



STEP 3: Set up the Master Node:

ssh into the master node, and download "**git**" and then clone the repository at:

https://github.com/Aadityaa2004/Financial-Investment-Risk-Analysis.git

And run "bash scripts/setup-master.sh". This script does:

1. Installs Docker on the instance
2. Adds the correct entries to /etc/hosts so the hostname master, worker1, worker2 resolve correctly
3. Generates an SSH key pair for Hadoop's internal communication
4. Builds the Docker images (downloads Hadoop 3.3.6, Go, and Java)
5. Starts the master container (HDFS NameNode, YARN, and 4 Go services)

will install Docker, configure hosts, and build images automatically.

```
ubuntu@ip-172-31-74-186:~/hadoop-risk-analysis$ sudo -E bash scripts/setup-master.sh
[setup-master] Checking prerequisites...
[setup-master] Docker already installed: Docker version 29.4.1, build 055a478
[setup-master] Docker Compose: Docker Compose version v5.1.3
[setup-master] Configuring /etc/hosts...
[setup-master]   Already present: 172.31.74.186 master
[setup-master]   Already present: 172.31.73.74 worker1
[setup-master]   Already present: 172.31.64.219 worker2
[setup-master] Setting up SSH keys...
[setup-master] Building Docker images (this takes ~10 minutes on first run)...
#18 naming to docker.io/library/hadoop-base:latest
#18 naming to docker.io/library/hadoop-base:latest done
#18 unpacking to docker.io/library/hadoop-base:latest
#18 unpacking to docker.io/library/hadoop-base:latest 26.6s done
#18 DONE 98.5s
#17 exporting manifest list sha256:7a82f57c948bb6c17652d8e3c4182a08c2390dc36e51a0228a9fe974932df95a 0.0s done
#17 naming to docker.io/library/hadoop-master:latest done
#17 unpacking to docker.io/library/hadoop-master:latest
#17 unpacking to docker.io/library/hadoop-master:latest 8.5s done
#17 DONE 37.8s
[setup-master] Starting master node containers...
WARN[0000] /home/ubuntu/hadoop-risk-analysis/docker-compose.master.yml: the attribute `version` is obsolete, it will be ignored, ple
ase remove it to avoid potential confusion
[+] up 4/4
 ✔ Volume hadoop-risk-analysis_hadoop-logs   Created                                                                    0.0s
 ✔ Volume hadoop-risk-analysis_namenode-data Created                                                                    0.0s
 ✔ Volume hadoop-risk-analysis_hadoop-tmp    Created                                                                    0.0s
 ✔ Container hadoop-master                   Started                                                                    0.5s
```

STEP 3: Set up the Worker Nodes:

Repeat the same process as STEP 3 and run "bash scripts/setup-worker.sh".

This script does:

1. Installs Docker
2. Adds /etc/hosts entries
3. Adds the master's SSH public key to authorized_keys (so Hadoop can connect without passwords)
4. Builds the worker Docker image
5. Starts the worker container (HDFS DataNode, and YARN NodeManager)
6. The worker automatically registers itself with the master's NameNode

```
ubuntu@ip-172-31-73-74:~/hadoop-risk-analysis$ sudo -E WORKER_ID=worker1 bash scripts/setup-worker.sh
[setup-worker] Setting up worker1...
[setup-worker] Docker already installed.
[setup-worker] Configuring /etc/hosts...
[setup-worker]   Added: 172.31.74.186 master
[setup-worker]   Added: 172.31.73.74 worker1
[setup-worker]   Added: 172.31.64.219 worker2
[setup-worker] Adding master SSH public key...
[setup-worker]   Master key added.
[setup-worker] Building worker Docker image...
#18 exporting manifest list sha256:d4d04f4337fbba6d445d7119b8ae4df78aa3e324fe001d22a12d998b0af7606b 0.0s done
#18 naming to docker.io/library/hadoop-base:latest done
#18 unpacking to docker.io/library/hadoop-base:latest
#18 unpacking to docker.io/library/hadoop-base:latest 24.4s done
#18 DONE 94.1s
#11 naming to docker.io/library/hadoop-worker:latest
#11 naming to docker.io/library/hadoop-worker:latest done
#11 unpacking to docker.io/library/hadoop-worker:latest
#11 unpacking to docker.io/library/hadoop-worker:latest 0.3s done
#11 DONE 1.0s
[setup-worker] Starting worker containers...
WARN[0000] /home/ubuntu/hadoop-risk-analysis/docker-compose.worker.yml: the attribute `version` is obsolete, it will be ignored, ple
ase remove it to avoid potential confusion
[+] up 4/4
 ✔ Volume hadoop-risk-analysis_hadoop-logs   Created                                                                    0.0s
 ✔ Volume hadoop-risk-analysis_datanode-data Created                                                                    0.0s
 ✔ Volume hadoop-risk-analysis_hadoop-tmp    Created                                                                    0.0s
 ✔ Container hadoop-worker                   Started                                                                    0.4s
```

STEP 4: Verify the Cluster Is Connected:

Go to the Master Node's terminal and run "docker exec hadoop-master hdfs dfsadmin -report". This will show all the connected nodes.

```
ubuntu@ip-172-31-74-186:~/hadoop-risk-analysis$ sudo docker exec hadoop-master hdfs dfsadmin -report
Configured Capacity: 62048567296 (57.79 GB)
Present Capacity: 46408724480 (43.22 GB)
DFS Remaining: 46408671232 (43.22 GB)
DFS Used: 53248 (52 KB)
DFS Used%: 0.00%
Replicated Blocks:
        Under replicated blocks: 0
        Blocks with corrupt replicas: 0
        Missing blocks: 0
        Missing blocks (with replication factor 1): 0
        Low redundancy blocks with highest priority to recover: 0
        Pending deletion blocks: 0
Erasure Coded Block Groups:
        Low redundancy block groups: 0
        Block groups with corrupt internal blocks: 0
        Missing block groups: 0
        Low redundancy blocks with highest priority to recover: 0
        Pending deletion blocks: 0

-------------------------------------------------
Live datanodes (2):

Name: 172.31.64.219:9866 (worker2)
Hostname: worker2
Decommission Status : Normal
Configured Capacity: 31024283648 (28.89 GB)
DFS Used: 24576 (24 KB)
Non DFS Used: 7805186048 (7.27 GB)
DFS Remaining: 23202295808 (21.61 GB)
DFS Used%: 0.00%
DFS Remaining%: 74.79%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sun Apr 26 22:18:39 GMT 2026
Last Block Report: Sun Apr 26 22:15:33 GMT 2026
Num of Blocks: 0


Name: 172.31.73.74:9866 (worker1)
Hostname: worker1
Decommission Status : Normal
Configured Capacity: 31024283648 (28.89 GB)
DFS Used: 28672 (28 KB)
Non DFS Used: 7801102336 (7.27 GB)
DFS Remaining: 23206375424 (21.61 GB)
DFS Used%: 0.00%
DFS Remaining%: 74.80%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sun Apr 26 22:18:40 GMT 2026
Last Block Report: Sun Apr 26 22:03:53 GMT 2026
Num of Blocks: 0
```

This returns Live Datanode as 2 which refers to the two workers that are connected to the Master Node.

STEP 5: Run the MapReduce Jobs:

Once the Connections are verified, you can run the map reduce task through running “bash scripts/run-all-jobs.sh”.

The process begins by uploading a large Kaggle CSV file to the HDFS input directory. This data is then processed by four distinct MapReduce jobs (Job 1 through Job 4), each consisting of a mapping and reducing function.

```
ubuntu@ip-172-31-74-186:~/hadoop-risk-analysis$ bash scripts/run-all-jobs.sh
[run-jobs] hadoop command not found. Is Hadoop installed and in PATH?
ubuntu@ip-172-31-74-186:~/hadoop-risk-analysis$ sudo docker exec hadoop-master hdfs dfs -ls -h /user/hadoop/lendingclub/input/
Found 1 items
-rw-r--r--   2 root supergroup      1.6 G 2026-04-26 22:38 /user/hadoop/lendingclub/input/accepted_2007_to_2018Q4.csv
ubuntu@ip-172-31-74-186:~/hadoop-risk-analysis$ sudo docker exec -it hadoop-master bash -lc '
export PATH=/opt/hadoop/bin:/opt/hadoop/sbin:$PATH
export HADOOP_HOME=/opt/hadoop
export HDFS_INPUT=/user/hadoop/lendingclub/input
export HDFS_OUTPUT_BASE=/user/hadoop/lendingclub/output

bash /mapreduce/job1-default-by-grade/run.sh
bash /mapreduce/job2-default-by-state/run.sh
bash /mapreduce/job3-default-by-employment/run.sh
bash /mapreduce/job4-interest-vs-default/run.sh
'
[job1] Removing stale output (if any)...
[job1] Submitting Default Rate by Loan Grade job...
2026-04-26 22:41:40,893 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [/mapreduce/job1-default-by-grade/mapper.py, /mapreduce/job1-default-by-grade/reducer.py, /tmp/hadoop-unjar3032370305981859907/] [] /tmp/streamjob1806092967706170029 6.jar tmpDir=null
2026-04-26 22:41:42,457 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at master/172.31.74.186:8032
2026-04-26 22:41:42,810 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at master/172.31.74.186:8032
2026-04-26 22:41:44,507 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_1777239736683_0001
2026-04-26 22:41:45,276 INFO mapred.FileInputFormat: Total input files to process : 1
2026-04-26 22:41:45,363 INFO mapreduce.JobSubmitter: number of splits:13
2026-04-26 22:41:46,190 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777239736683_0001
2026-04-26 22:41:46,190 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-26 22:41:46,455 INFO conf.Configuration: resource-types.xml not found
2026-04-26 22:41:46,456 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-04-26 22:41:48,888 INFO impl.YarnClientImpl: Submitted application application_1777239736683_0001
2026-04-26 22:41:48,959 INFO mapreduce.Job: The url to track the job: http://master:8088/proxy/application_1777239736683_0001/
2026-04-26 22:41:48,960 INFO mapreduce.Job: Running job: job_1777239736683_0001
2026-04-26 22:42:02,259 INFO mapreduce.Job: Job job_1777239736683_0001 running in uber mode : false
2026-04-26 22:42:02,260 INFO mapreduce.Job:  map 0% reduce 0%
2026-04-26 22:42:13,540 INFO mapreduce.Job:  map 8% reduce 0%
2026-04-26 22:42:18,584 INFO mapreduce.Job: Task Id : attempt_1777239736683_0001_m_000002_0, Status : FAILED
[2026-04-26 22:42:17.630]Container killed on request. Exit code is 137
[2026-04-26 22:42:17.637]Container exited with a non-zero exit code 137.
[2026-04-26 22:42:17.645]Killed by external signal

2026-04-26 22:42:23,654 INFO mapreduce.Job:  map 15% reduce 0%
2026-04-26 22:42:29,708 INFO mapreduce.Job:  map 25% reduce 0%
2026-04-26 22:42:30,715 INFO mapreduce.Job:  map 31% reduce 0%
2026-04-26 22:42:33,740 INFO mapreduce.Job:  map 38% reduce 0%
2026-04-26 22:42:42,804 INFO mapreduce.Job:  map 46% reduce 0%
2026-04-26 22:42:50,844 INFO mapreduce.Job:  map 46% reduce 15%
2026-04-26 22:42:53,861 INFO mapreduce.Job: Task Id : attempt_1777239736683_0001_m_000007_0, Status : FAILED
[2026-04-26 22:42:52.171]Container killed on request. Exit code is 137
[2026-04-26 22:42:52.171]Container exited with a non-zero exit code 137.
[2026-04-26 22:42:52.173]Killed by external signal

2026-04-26 22:42:54,867 INFO mapreduce.Job:  map 51% reduce 15%
2026-04-26 22:42:55,872 INFO mapreduce.Job:  map 54% reduce 15%
2026-04-26 22:43:02,925 INFO mapreduce.Job:  map 54% reduce 18%
2026-04-26 22:43:03,932 INFO mapreduce.Job:  map 59% reduce 18%
2026-04-26 22:43:04,938 INFO mapreduce.Job:  map 62% reduce 18%
2026-04-26 22:43:08,963 INFO mapreduce.Job:  map 62% reduce 21%
2026-04-26 22:43:17,002 INFO mapreduce.Job: Task Id : attempt_1777239736683_0001_m_000009_0, Status : FAILED
[2026-04-26 22:43:16.113]Container killed on request. Exit code is 137
[2026-04-26 22:43:16.116]Container exited with a non-zero exit code 137.
[2026-04-26 22:43:16.118]Killed by external signal

2026-04-26 22:43:18,008 INFO mapreduce.Job:  map 66% reduce 21%
2026-04-26 22:43:20,018 INFO mapreduce.Job:  map 69% reduce 21%
2026-04-26 22:43:21,023 INFO mapreduce.Job:  map 69% reduce 23%
2026-04-26 22:43:27,050 INFO mapreduce.Job:  map 74% reduce 23%
2026-04-26 22:43:29,059 INFO mapreduce.Job:  map 77% reduce 23%
2026-04-26 22:43:33,079 INFO mapreduce.Job:  map 77% reduce 26%
2026-04-26 22:43:41,113 INFO mapreduce.Job:  map 81% reduce 26%
2026-04-26 22:43:41,115 INFO mapreduce.Job: Task Id : attempt_1777239736683_0001_m_000009_1, Status : FAILED
[2026-04-26 22:43:39.683]Container killed on request. Exit code is 137
[2026-04-26 22:43:39.687]Container exited with a non-zero exit code 137.
```

STEP 6: Obtain Results:

Once this is done, You can view Individual Results by running the following commands:

1) curl http://54.210.15.33:8080/api/results/grade:

```
ubuntu@ip-172-31-74-186:~/hadoop-risk-analysis$ curl http://master:8080/api/results/grade
[{"grade":"A","total_loans":235193,"total_defaults":14214,"default_rate_pct":6.04,"risk_leve
l":"MEDIUM"},{"grade":"B","total_loans":393102,"total_defaults":52661,"default_rate_pct":13.
4,"risk_level":"MEDIUM"},{"grade":"C","total_loans":382323,"total_defaults":85805,"default_r
ate_pct":22.44,"risk_level":"HIGH"},{"grade":"D","total_loans":201657,"total_defaults":61264
,"default_rate_pct":30.38,"risk_level":"CRITICAL"},{"grade":"E","total_loans":94192,"total_d
efaults":36199,"default_rate_pct":38.43,"risk_level":"CRITICAL"},{"grade":"F","total_loans":
32306,"total_defaults":14585,"default_rate_pct":45.15,"risk_level":"CRITICAL"},{"grade":"G",
"total_loans":9326,"total_defaults":4632,"default_rate_pct":49.67,"risk_level":"CRITICAL"}]
```

This table details loan default rates categorized by loan grade:

Grade	Total Loans	Total Defaults	Default Rate (%)	Risk Level
A	235,193	14,214	6.04	MEDIUM
B	393,102	52,661	13.40	MEDIUM
C	382,323	85,805	22.44	HIGH
D	201,657	61,264	30.38	CRITICAL
E	94,192	36,199	38.43	CRITICAL
F	32,306	14,585	45.15	CRITICAL
G	9,326	4,632	49.67	CRITICAL

2) curl http://54.210.15.33:8080/api/results/state:

```
ubuntu@ip-172-31-74-186:~/hadoop-risk-analysis$ curl http://master:8080/api/results/state
[{"state":"AK","total_loans":3195,"total_defaults":628,"default_rate_pct":19.66,"risk_level"
:"HIGH"},{"state":"AL","total_loans":16645,"total_defaults":3934,"default_rate_pct":23.63,"r
isk_level":"HIGH"},{"state":"AR","total_loans":10062,"total_defaults":2426,"default_rate_pct
":24.11,"risk_level":"HIGH"},{"state":"AZ","total_loans":32748,"total_defaults":6438,"defaul
t_rate_pct":19.66,"risk_level":"HIGH"},{"state":"CA","total_loans":196853,"total_defaults":3
8642,"default_rate_pct":19.63,"risk_level":"HIGH"},{"state":"CO","total_loans":29736,"total_
defaults":4620,"default_rate_pct":15.54,"risk_level":"HIGH"},{"state":"CT","total_loans":197
91,"total_defaults":3441,"default_rate_pct":17.39,"risk_level":"HIGH"},{"state":"DC","total_
loans":3485,"total_defaults":461,"default_rate_pct":13.23,"risk_level":"MEDIUM"},{"state":"D
E","total_loans":3805,"total_defaults":751,"default_rate_pct":19.74,"risk_level":"HIGH"},{"s
tate":"FL","total_loans":95843,"total_defaults":20608,"default_rate_pct":21.5,"risk_level":"
HIGH"},{"state":"GA","total_loans":43480,"total_defaults":8013,"default_rate_pct":18.43,"ris
k_level":"HIGH"},{"state":"HI","total_loans":6765,"total_defaults":1368,"default_rate_pct":2
0.22,"risk_level":"HIGH"},{"state":"IA","total_loans":14,"total_defaults":3,"default_rate_pc
t":21.43,"risk_level":"HIGH"},{"state":"ID","total_loans":1692,"total_defaults":318,"default
_rate_pct":18.79,"risk_level":"HIGH"},{"state":"IL","total_loans":51870,"total_defaults":939
9,"default_rate_pct":18.12,"risk_level":"HIGH"},{"state":"IN","total_loans":21726,"total_def
aults":4656,"default_rate_pct":21.43,"risk_level":"HIGH"},{"state":"KS","total_loans":11267,
"total_defaults":1888,"default_rate_pct":16.76,"risk_level":"HIGH"},{"state":"KY","total_loa
ns":12871,"total_defaults":2704,"default_rate_pct":21.01,"risk_level":"HIGH"},{"state":"LA",
"total_loans":15524,"total_defaults":3598,"default_rate_pct":23.18,"risk_level":"HIGH"},{"st
ate":"MA","total_loans":31071,"total_defaults":5926,"default_rate_pct":19.07,"risk_level":"H
IGH"},{"state":"MD","total_loans":31298,"total_defaults":6678,"default_rate_pct":21.34,"risk
_level":"HIGH"},{"state":"ME","total_loans":2030,"total_defaults":281,"default_rate_pct":13.
84,"risk_level":"MEDIUM"},{"state":"MI","total_loans":35308,"total_defaults":7170,"default_r
ate_pct":20.31,"risk_level":"HIGH"},{"state":"MN","total_loans":24004,"total_defaults":4743,
"default_rate_pct":19.76,"risk_level":"HIGH"},{"state":"MO","total_loans":21340,"total_defau
```

The analysis shows that the majority of states fall under the HIGH risk level. The states with CRITICAL and MEDIUM risk classifications are detailed below:

- **CRITICAL Risk States**:
  - MS (Mississippi): 26.11% default rate
  - NE (Nebraska): 25.22% default rate
- **MEDIUM Risk States**:
  - DC: 13.23% default rate
  - ME (Maine): 13.84% default rate
  - VT (Vermont): 13.97% default rate
  - OR (Oregon): 14.43% default rate
  - NH (New Hampshire): 14.57% default rate

3) curl http://54.210.15.33:8080/api/results/employment

```
ubuntu@ip-172-31-74-186:~/hadoop-risk-analysis$ curl http://master:8080/api/results/employme
nt
[{"employment_bucket":"1-2 years","total_loans":210943,"total_defaults":42495,"default_rate_
pct":20.15,"risk_level":"HIGH"},{"employment_bucket":"10+ years","total_loans":442679,"total
_defaults":83223,"default_rate_pct":18.8,"risk_level":"HIGH"},{"employment_bucket":"3-5 year
s","total_loans":272957,"total_defaults":54066,"default_rate_pct":19.81,"risk_level":"HIGH"}
,{"employment_bucket":"6-9 years","total_loans":234433,"total_defaults":46130,"default_rate_
pct":19.68,"risk_level":"HIGH"},{"employment_bucket":"\u003c 1 year","total_loans":108537,"t
otal_defaults":22296,"default_rate_pct":20.54,"risk_level":"HIGH"}]
```

Employment Bucket	Total Loans	Total Defaults	Default Rate (%)	Risk Level
10+ years	442,679	83,223	18.80	HIGH
6-9 years	234,433	46,130	19.68	HIGH
3-5 years	272,957	54,066	19.81	HIGH
1-2 years	210,943	42,495	20.15	HIGH
< 1 year	108,537	22,296	20.54	HIGH

4) curl http://54.210.15.33:8080/api/results/interest

```
ubuntu@ip-172-31-74-186:~/hadoop-risk-analysis$ curl http://master:8080/api/results/interest
[{"grade":"A","total_loans":235193,"avg_interest_rate":7.11,"default_rate_pct":6.04,"risk_le
vel":"MEDIUM"},{"grade":"B","total_loans":393102,"avg_interest_rate":10.68,"default_rate_pct
":13.4,"risk_level":"MEDIUM"},{"grade":"C","total_loans":382323,"avg_interest_rate":14.02,"d
efault_rate_pct":22.44,"risk_level":"HIGH"},{"grade":"D","total_loans":201657,"avg_interest_
rate":17.71,"default_rate_pct":30.38,"risk_level":"CRITICAL"},{"grade":"E","total_loans":941
92,"avg_interest_rate":21.11,"default_rate_pct":38.43,"risk_level":"CRITICAL"},{"grade":"F",
"total_loans":32306,"avg_interest_rate":24.88,"default_rate_pct":45.15,"risk_level":"CRITICA
L"},{"grade":"G","total_loans":9326,"avg_interest_rate":27.54,"default_rate_pct":49.67,"risk
_level":"CRITICAL"}]
```

Grade	Total Loans	Avg Interest Rate (%)	Default Rate (%)	Risk Level
A	235,193	7.11	6.04	MEDIUM
B	393,102	10.68	13.40	MEDIUM
C	382,323	14.02	22.44	HIGH
D	201,657	17.71	30.38	CRITICAL
E	94,192	21.11	38.43	CRITICAL
F	32,306	24.88	45.15	CRITICAL
G	9,326	27.54	49.67	CRITICAL

5) curl http://54.210.15.33:8080/api/results/risk-summary

```
ubuntu@ip-172-31-74-186:~/hadoop-risk-analysis$ curl http://master:8080/api/results/risk-sum
mary
{"highest_risk_grade":"G","highest_risk_state":"MS","highest_risk_employment_bucket":"\u003c
 1 year","lowest_risk_grade":"A","overall_default_rate_pct":19.980728418313493,"total_loans_
analyzed":1348099,"recommendation":"Based on 1348099 loans analyzed: avoid Grade G loans (49
.7% default rate). Grade A loans offer the best risk-adjusted returns. Borrowers with \u003c
 1 year employment history carry the highest default risk."}
ubuntu@ip-172-31-74-186:~/hadoop-risk-analysis$
```

This summary aggregates the key risk findings from the 1,348,099 loans analyzed:

- **Total Loans Analyzed**: 1,348,099
- **Overall Default Rate**: 19.98%
- **Lowest Risk Grade**: A
- **Highest Risk Grade**: G
- **Highest Risk State**: MS
- **Highest Risk Employment Bucket**: < 1 year
- **Recommendation**: Based on 1,348,099 loans analyzed: avoid Grade G loans (49.7% default rate). Grade A loans offer the best risk-adjusted returns. Borrowers with < 1 year employment history carry the highest default risk.